Automated Revenue Replacement and Treasury Balancing Mechanism

Proposal for meeting cash shortfall in West African Countries – Federal Republic of Nigeria as a Case Study

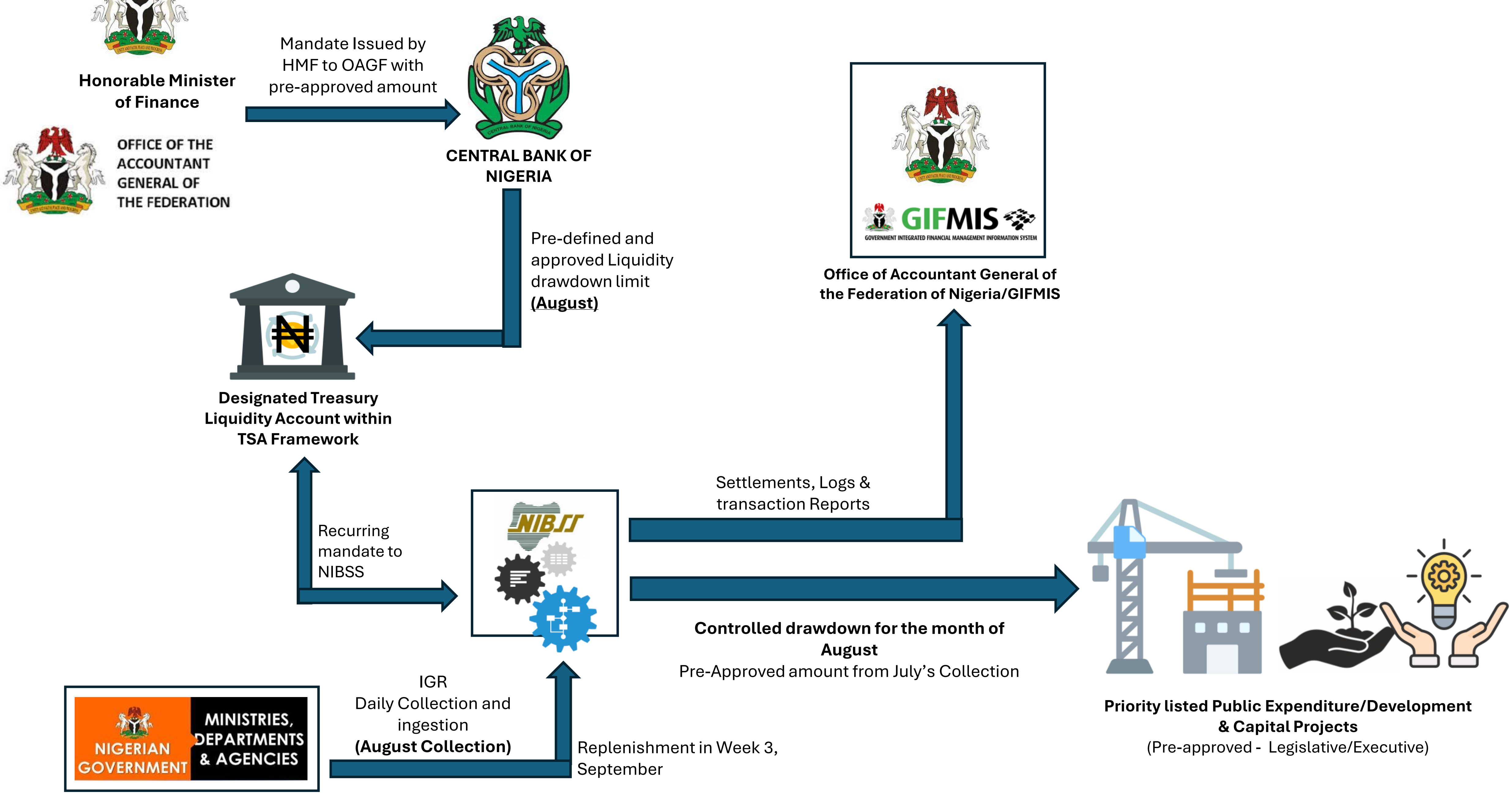
Key Assumptions

- **Total IGR collected in month** (example): ₦2,000,000,000,000 (Aug).
- **Monthly liquidity facility / drawdown cap**: ₦300,000,000,000.
- **Typical remittance timing**: bulk inflows arrive in the **3rd week of the following month** (i.e., Sept 3rd week covers August).

Month-to-Month Relationship

Month's Drawdown (₦300bn)	Replenished By IGR Collected In	Replenishment Happens In (3rd Week Of)
August Drawdown	August IGR	**September**
September Drawdown	September IGR	**October**
October Drawdown	October IGR	**November**

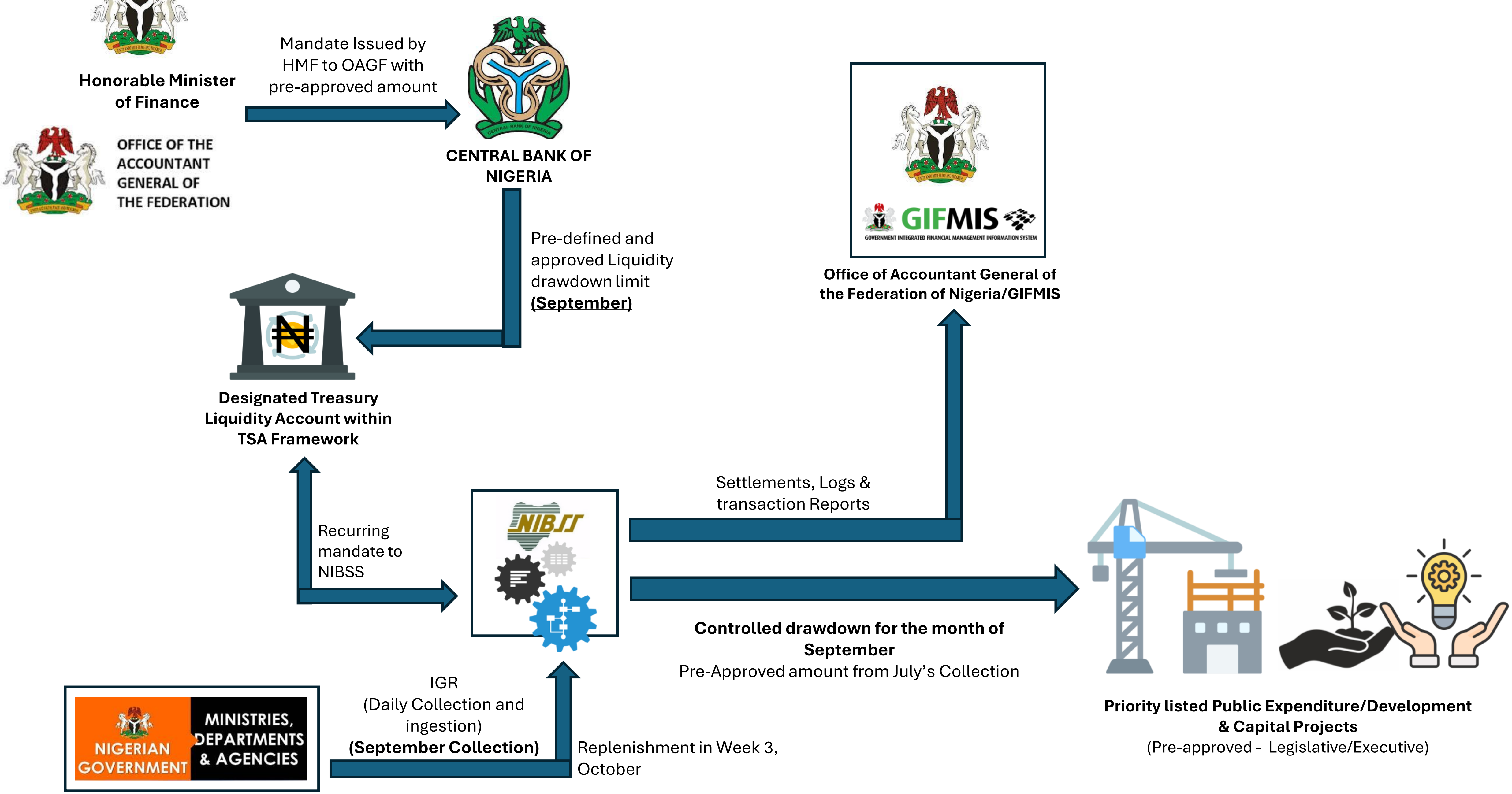
1st – 31st August 2025 inflow



AUGUST – First Drawdown (₦300bn Liquidity Advance)

PERIOD	EVENT	EXPLANATION
Start of August (Week 1)	₦300bn drawdown from Designated Treasury Liquidity Account (DTLA)	The OAGF/CBN releases ₦300bn to fund priority capital or expenditure projects.
Throughout August	MDAs collect IGR (≈ ₦2 trillion)	Actual inflows accumulate in commercial banks and TSA-linked accounts.
End of August	Month closes	₦300bn used; ₦2tn IGR accrued but not yet fully received at CBN.

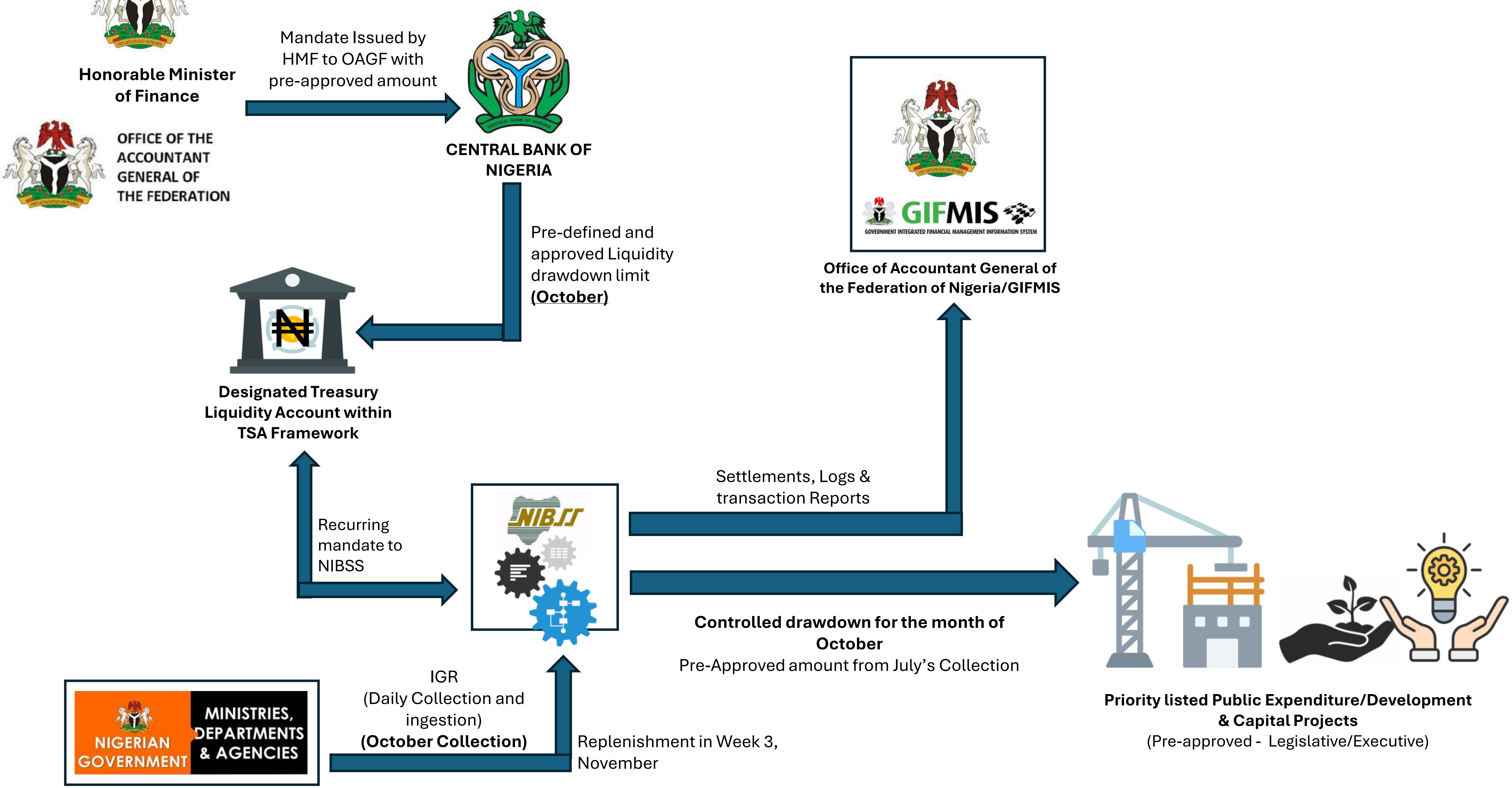
1st – 30th September 2025 inflow



September – First Replacement And Second Drawdown

PERIOD	EVENT	EXPLANATION
3rd Week of September	Bulk IGR inflows (for August) hit the TSA (≈ ₦2tn)	ARSS system reconciles August IGR inflows and sweeps ₦300bn equivalent to replenish August drawdown.
Result	August ₦300bn drawdown fully replaced	The Treasury position is restored automatically by ARSS.
Start of September (Week 1)	New ₦300bn drawdown for September projects	OAGF/CBN authorizes the next tranche for new project expenditures.
Throughout September	MDAs collect September IGR (≈ ₦2 trillion expected)	These inflows will reach the TSA in the **3rd week of October**.
End of September	ARSS monthly reconciliation	August replenishment confirmed, September drawdown open, and predictive analytics prepared for October inflows.

1st – 31st October 2025 inflow



October – Second Replacement And Third Drawdown

PERIOD	EVENT	EXPLANATION
3rd Week of October	Bulk IGR inflows for **September** reach the TSA (~₦2tn)	ARSS automatically reconciles inflows and sweeps ₦300bn to cover the September drawdown.
Result	September ₦300bn fully replenished	Treasury balance restored again; rolling liquidity cycle maintained.
Start of October (Week 1)	New ₦300bn drawdown approved for October project disbursement	OAGF/CBN initiates fresh liquidity for the month’s capital expenditure.
Throughout October	MDAs collect IGR (≈ ₦2 trillion expected)	Actual remittances are delayed and come bulk in **November (3rd week)**.
End of October	ARSS generates reconciliation summary	Shows three sequential cycles: August replenished in September, September replenished in October, and October drawdown open.

November – Third Replacement and Fourth Drawdown (Continuity Of Cycle)

PERIOD	EVENT	EXPLANATION
3rd Week of November	Bulk IGR inflows for **October** reach TSA (~₦2tn)	ARSS reconciles inflows and replenishes ₦300bn drawn at start of October.
Result	October ₦300bn drawdown replenished	The treasury is back in equilibrium — the revolving mechanism continues.
Start of November (Week 1)	New ₦300bn drawdown for November projects	OAGF/CBN continues the liquidity cycle for next month's expenditure commitments.
Throughout November	MDAs collect IGR for November (≈ ₦2tn expected)	Will be remitted and sweep-triggered in **3rd week of December**.
End of November	ARSS reconciles October replenishment and November drawdown status	Reports generated and archived for audit.

Risk Assessment Matrix

RISK CATEGORY	NATURE	IMPACT LEVEL	MITIGATION STRATEGY
Timing mismatch	Operational	High	Predictive monitoring, contingency reserve
Incomplete MDA remittances	Structural	High	Data harmonization, enforcement mechanism
IGR variability	Fiscal	Medium	Dynamic drawdown scaling
Automation/system failure	Technical	High	Security hardening, redundancy
Coordination gaps	Governance	Medium	Joint Treasury Task Force
Legal ambiguity	Regulatory	Medium	Implementation Circular
Liquidity concentration	Financial	Medium	Short-term investment mechanism
Data errors	Technical	Medium	Multi-source verification
Change resistance	Human	Medium	Structured change management
Political interference	Institutional	Medium	Audit logging and transparency reports

Challenges & Mitigation Steps

1. <u>Timing Mismatch Between IGR Accrual and Actual Inflows</u>

Description:

While the model assumes that IGR inflows consistently arrive in the **third week of each month**, the actual timing of receipts may vary due to administrative delays, late remittances by MDAs, or issues in interbank settlement processes.

Impact:

A delay of even one week could cause a temporary **cash flow gap** for the replacement of the ₦300 billion drawdown, creating liquidity strain at the CBN’s consolidated position.

Mitigation:

- Implement predictive inflow monitoring using **historical trend analytics** and **variance alerts**.
- Maintain a **10% contingency reserve buffer** within the Treasury Single Account (TSA) liquidity framework.
- Integrate **real-time payment triggers** via the ARSS platform with alerts to OAGF Treasury Operations

Challenges & Mitigation Steps/2

2. <u>Incomplete or Non-Uniform IGR Remittances by MDAs</u>

Description:

Not all Ministries, Departments, and Agencies (MDAs) remit IGR promptly or uniformly. Certain revenue lines (e.g., petroleum royalties, port fees, customs duties) have variable collection cycles.

Impact:

Partial or delayed remittances distort total inflow assumptions, leading to under-coverage of the previous month’s drawdown.

Mitigation:

- Enforce **mandatory TSA remittance timelines** with penalties for delay.
- Integrate **automatic reconciliation of MDA revenue schedules** with the ARSS.
- Deploy **data harmonization with NIBSS and CBN payment systems** to detect shortfalls in real time

Challenges & Mitigation Steps/3

3. <u>Over-Dependence on Predictable IGR Volume</u>

Description:

The model assumes an average monthly IGR inflow of about **₦2 Trillion**. However, macroeconomic shocks, oil price volatility, tax seasonality, or policy changes could reduce collections below expected thresholds.

Impact:

Lower-than-expected IGR could jeopardize the replenishment cycle, leading to extended float or forced borrowing.

Mitigation:

- Introduce **dynamic scaling** — adjust drawdown size monthly based on actual IGR trend analysis.
- Include **early-warning IGR variance metrics** in ARSS dashboards.
- Maintain **CBN Treasury buffer instruments** (e.g., repo facilities or short-term notes) for emergency coverage.

Challenges & Mitigation Steps/4

4. Systemic and Operational Risk in Automation

Description:

An automated system that pulls bank statements, reconciles TSA balances, and executes fund sweeps carries inherent IT and cybersecurity risks.

Impact:

Failure of the ARSS infrastructure or unauthorized access could compromise fund movement accuracy or expose financial data.

Mitigation:

- Deploy **tiered user authentication and rule-based transaction approval**.
- Host ARSS on a **CBN-approved secure cloud or hybrid data center** environment with **redundant backup (Provide an interface for OAGF and Ministry of Finance**.
- Implement **continuous penetration testing, audit logs, and role-based encryption**.

Challenges & Mitigation Steps/5

5. Inter-Agency Coordination Challenges

Description:

The model requires close operational synchronization between **OAGF**, **CBN**, **FIRS**, **NIBSS**, and **commercial banks** hosting TSA-linked accounts.

Impact:

Poor coordination could result in fragmented data, inconsistent reconciliation, and delays in sweep operations.

Mitigation:

- Establish a **Joint Treasury Implementation Task Force (JTITF)** chaired by OAGF.
- Adopt a secure **standardized data exchange formats (API-based)** across all participating institutions.
- Introduce **Service-Level Agreements (SLAs)** with participating banks for daily data delivery.

Challenges & Mitigation Steps/6

6. Policy and Legal Framework Constraints

Description:

Current Public Financial Management (PFM) regulations and TSA operational guidelines may not explicitly support automated intra-month drawdowns and replenishments.

Impact:

Absence of enabling circulars or directives could delay implementation or create compliance uncertainty.

Mitigation:

- Secure **formal approval from the Fiscal Responsibility Department and the Debt Management Office (DMO)**.
- Draft an **OAGF/CBN Implementation Circular** defining the legal standing of automated intra-month liquidity management.
- Ensure alignment with the **Fiscal Responsibility Act (FRA)** and **Financial Regulations 2009 (as amended)**.

Challenges & Mitigation Steps/7

7. Liquidity Concentration and Interest Opportunity Cost

Description:

Maintaining a ₦300 billion idle float (even temporarily) in a single CBN account may attract **opportunity cost** versus potential yield if invested in short-term instruments.

Impact:

Lost earnings potential or inefficiency in government liquidity utilization.

Mitigation:

- Structure the ₦300 billion as a **revolving liquidity pool** that can be invested overnight in **risk-free OMO bills** or **treasury placements** pending utilization.
- Automate the investment and recall process through **CBN Treasury Systems Integration (TSI)**.

Challenges & Mitigation Steps/8

8. Data Integrity and Reconciliation Errors

Description:

Automated statement pulling and reconciliation from multiple bank sources (CBN, commercial banks, TSA sub-accounts) introduces risk of **data inconsistency or duplication**.

Impact:

Erroneous balances could trigger wrong sweep values or delayed settlement.

Mitigation:

- Utilize **checksum validation** and **multi-source verification** before each monthly reconciliation.
- Include **AI-driven anomaly detection** within ARSS to flag outliers or inconsistent entries.
- Enforce **four-eye principle** (automated + human verification) before fund settlement.

Challenges & Mitigation Steps/9

9. Institutional Change Management

Description:

Adopting automated liquidity management and predictive analytics represents a **major operational and cultural shift** for OAGF and CBN staff.

Impact:

Resistance to automation, lack of user adoption, and manual overrides could undermine efficiency.

Mitigation:

- Develop a **change management and capacity-building plan** (training, process redefinition, workflow mapping).
- Conduct **joint OAGF–CBN–CAO Trust/Upperlink workshops** to demonstrate transparency and control improvements.
- Deploy **user-friendly dashboards and stepwise onboarding protocols** to improve acceptance.

Challenges & Mitigation Steps/10

10. Political or Fiscal Policy Interference

Description:

Sudden political directives to alter drawdown amounts or suspend TSA sweeps could disrupt the automated process.

Impact:

Inconsistency in implementation, reduced fiscal discipline, or reputational risk to participating institutions.

Mitigation:

- Embed **policy override logging** in the ARSS for audit visibility.
- Maintain **communication protocol between OAGF, CBN, and the Presidency** for exceptional interventions.
- Publish **monthly ARSS Transparency Reports** to sustain public and institutional accountability.

Conclusion

- The ₦300bn fund acts as a **rolling liquidity window**, not an additional expenditure — it is continuously replaced by subsequent IGR inflows.
- There is always **one open drawdown** in the system (the current month) and **one replenishment** occurring (for the prior month).
- The mechanism ensures:
  - **Drawdown automation and approval control** (early month)
  - **Replenishment automation** (mid-next month)
  - **Reconciliation and audit trail generation** (month-end)
- Fiscal sustainability is preserved because no permanent overdraft or new borrowing occurs.